

Family 0.9.tf in two pages

family 1.0.tf with its raw imagery replaced by Google’s AlphaEarth embedding

blauewelt/earth — two-page summary of the design note, 20 September 2026. A glossary of every term follows the two pages.

What it is. Family 0.9.tf is family 1.0.tf (the land-and-coast stores) with **one substitution:** instead of the list of raw satellite and aerial images — and the small learned encoder that would turn them into tokens — it uses Google DeepMind’s published **AlphaEarth Satellite Embedding**: 64 numbers describing each 10 m of land, once a year, 2017–2025, computed from Sentinel-1, Sentinel-2 and Landsat and trained against radar, LiDAR, climate and land-cover sources; free under CC BY 4.0, 577 TB in all. **Everything else is inherited from 1.0.tf by reference** — the stations and gauges, the LiDAR, the daily fields, the prediction targets, the fixed maps, and the whole of 1.0.tf’s biosphere (canopy height and the canopy catalogues, fluorescence, leaf area, the leaf calendar, the species records) — exactly as family 10.2 inherits family 10.1: the registry lists them with their 1.0.tf locations and checksums, not one byte is rebuilt, and this family builds nothing new for the living surface. **What it is for:** testing land prediction now, and measuring exactly whether the programme’s own image encoder is worth its compute — the two families differ only in the imagery. The canopy is where that measurement is sharpest: the AlphaEarth embedding was itself trained against GEDI, so canopy height at held-out laser footprints is where this family’s cache and the programme’s own tile codec meet with everything else held fixed. **What it is not:** not sub-annual (one vector per year, so a five-day event is averaged away), not extensible to new sensors (NISAR) or to years before 2017, and not a copy of Google’s 577 TB, which is read in place.

store	what it is	producer	space	time	record	stored size	track	phase
<i>built by this family: the AlphaEarth Satellite Embedding, v1 (CC BY 4.0)</i>								
aef1k	embedding averaged over each km ² , plus how uniform the cell is	Google DeepMind (AlphaEarth v1); pooled here	0.93 km (1/120°)	annual	2017–2025	≈ 0.2 TB (0.25 TB with high latitudes)	public	A
aef.dots	16 unaveraged 10 m vectors per 0.25° land cell per year	Google DeepMind (AlphaEarth v1)	10 m points	annual	2017–2025	7 GB	public	A
aef100r	embedding averaged over 100 m cells, chosen regions	Google DeepMind (AlphaEarth v1); pooled here	93 m (1/1200°)	annual	2017–2025	≈ 0.35 TB (frontiers); ≈ 5.7 TB (coastal band, C)	public	B / C
aef10	Google’s original 10 m files, read in place by byte range	Google Cloud Storage; Source Cooperative mirror	10 m	annual	2017–2025	0 (577 TB referenced)	public (external)	A
<i>inherited from 1.0.tf by reference: same files, same checksums, nothing rebuilt</i>								
ghcnd	daily temperature, rain and snow at weather stations	NOAA NCEI (GHCN-Daily)	point; 132,437 stations	daily	1763 →	≈ 40 GB (probe: 41 B/row)	<i>inherited from 1.0.tf</i> ; public	A
igra	weather-balloon soundings of the air column	NOAA NCEI (IGRA v2.2)	point; 2,932 sites	00 and 12 UTC	1905 →	7–9 GB (probe: note 10 GB)	<i>inherited from 1.0.tf</i> ; public	A
tide	hourly coastal sea level from tide gauges	GESLA-4 via UHSLC	point; 6,474 records	hourly or finer	1800s → 2026	40–50 GB (probe: 33 B/row)	<i>inherited from 1.0.tf</i> ; public	A
tide.private	tide gauges whose terms forbid republishing	GESLA-4 (CMEMS, CV, UZ; SANHO)	point	hourly or finer	as tide	10–20 GB [†] (probe: 33 B/row)	<i>inherited from 1.0.tf</i> ; private	A
ndbc	weather and waves at US buoys and coastal stations	NOAA NDBC	point; 1,345 stations	hourly; 10 min recently	1970 → 2025	≈ 30 GB (probe: note 15–25 GB)	<i>inherited from 1.0.tf</i> ; public	A
gliders	ocean profiles from autonomous underwater gliders	IOOS Glider DAC; OceanGliders	point profile	every ≈ 3 h	2003 →	0.8 GB (probe: 155 B/row)	<i>inherited from 1.0.tf</i> ; public	A
flux	land–air exchange of CO ₂ , water and heat	FLUXNET (AmeriFlux, ICOS, TERN, ChinaFLUX)	point; 700+ towers	half-hourly	1991 →	5 GB	<i>inherited from 1.0.tf</i> ; public	A
fire	satellite detections of active fires	NASA FIRMS (MODIS, VIIRS)	1 km / 375 m	per overpass	2000–11 →	20 GB	<i>inherited from 1.0.tf</i> ; public	A
river	daily river discharge (US and Caravan)	USGS NWIS; GRDC-Caravan	point; 24,976 + 5,356	daily	1800s →	14 GB	<i>inherited from 1.0.tf</i> ; public	B
river.grdc	daily river discharge worldwide	GRDC	point; > 10,000 gauges	daily	1800s →	≈ 5 GB [†]	<i>inherited from 1.0.tf</i> ; private	B
ismn	soil moisture and temperature in the ground	ISMN (TU Wien)	point; 3,300 stations	hourly	1952 →	≈ 15 GB [†]	<i>inherited from 1.0.tf</i> ; private	B
ghcnh	hourly weather-station reports	NOAA NCEI (GHCNh)	point; 38,870 stations	hourly / synoptic	1700s →	360 GB	<i>inherited from 1.0.tf</i> ; public	B
hfr	surface currents from shore-based radar	HFRNet (US); European HFR Node	6 / 2 / 1 km cells	hourly	US 2012 →; EU ≈2009 →	110–155 GB	<i>inherited from 1.0.tf</i> ; public	B
alticap	satellite sea level close to the coast	ALTICAP (AVISO/ODATIS) Jason-3	≈ 350 m along track	≈ 10-day repeat	2016–02 → 2021–07	3 GB	<i>inherited from 1.0.tf</i> ; public	B

store	what it is	producer	space	time	record	stored size	track	phase
ecad	European station weather missing from GHCN-Daily	ECA&D (KNMI)	point	daily	1800s →	$\leq 20 \text{ GB}^\dagger$	<i>inherited from 1.0.tf</i> ; private	C
gedi	laser canopy height, cover and biomass	NASA GEDI (LP DAAC, ORNL)	25 m footprints	per orbit pass	2019-04 → 2023-03; 2024-04 → 2025-07	0.2–0.5 TB	<i>inherited from 1.0.tf</i> ; public	A (one year) / B
icesat2	laser terrain, canopy and sea-ice height	NASA ICESat-2 (NSIDC)	100 m segments ($\approx 11 \text{ m}$ spots)	per orbit pass	2018-10 →	$\approx 1.1 \text{ TB}$	<i>inherited from 1.0.tf</i> ; public	A (one year) / B
canopy30	how tall the canopy is, one map at 30 m	UMD GLAD (forest height 2020)	30 m	one year	2020	60–100 $\text{GB}^\dagger$ (261 tiles measured)	<i>inherited from 1.0.tf</i> ; public	A
canopy_ref	the finer canopy maps, listed not copied	ETH Zurich (10 m); Meta/WRI (1 m)	10 m and 1 m	annual	2020; 2018–2020	catalogue $< 1 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	A
sif	fluorescence: photosynthesis seen from orbit	TROPOMI (Caltech, ESA); NASA OCO-2, OCO-3	$7 \times 3.5 \text{ km}$; $1.3 \times 2.25 \text{ km}$	per sounding	2018-04 →; 2014-09 →	60–160 $\text{GB}^\dagger$	<i>inherited from 1.0.tf</i> ; public	A
lai500 (E7)	leaf area, and the light the canopy absorbs	NASA MODIS, VIIRS	500 m	8-day composite	2000-02 →	$\approx 60 \text{ GB}^\dagger$ per sensor-year	<i>inherited from 1.0.tf</i> ; public	A / B
pheno500	the leaf calendar: green-up, peak, dormancy	NASA MODIS (MCD12Q2)	500 m	annual	2001 → 2025	$\approx 10 \text{ GB}^\dagger$ a year	<i>inherited from 1.0.tf</i> ; public	A
gbif	dated, located records of species	GBIF snapshot (AWS Open Data)	point	per record	1600s →; 3.67 B measured	$\approx 90 \text{ GB}^\dagger$ public	<i>inherited from 1.0.tf</i> ; public + private	A
cat_biomass_esa	list of ESA’s P-band radar products	ESA BIOMASS (2025)	—	per product	2025 →	catalogue	<i>inherited from 1.0.tf</i> ; terms unverified	A if public
1st05	land-surface temperature, day and night (Terra in phase A)	NASA MODIS Terra (+ Aqua, VIIRS)	5.6 km (0.05°)	daily	2000-02 →	$\approx 350 \text{ GB}$ per satellite	<i>inherited from 1.0.tf</i> ; public	A / B
snow05	snow-cover fraction	NASA MODIS (NSIDC)	0.05°	daily	2000-02 →	$\approx 60 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	A
ref105	surface reflectance: the colour of the ground	NASA MODIS (+ VIIRS)	0.05°	daily	2000-02 →	$\approx 700 \text{ GB}$ (+330 GB VIIRS)	<i>inherited from 1.0.tf</i> ; public	B
ims1k	analyst-drawn snow and ice chart	NOAA IMS (NSIDC)	1 km (4 km and 24 km earlier)	daily	1 km 2014-12 →; 4 km 2004, 24 km 1997	$\approx 50 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	B
chirps05 (E4)	rainfall from satellite infrared and gauges	UCSB CHC (CHIRPS v3.0)	0.05°	pentad (6 a month)	1981 →	$\approx 40 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	B
alerts	dated forest-clearing alerts	UMD GLAD, WUR RADD, NASA OPERA	10–30 m	per detection	2015 →	$\approx 30 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	A
lossyear	year each 30 m pixel lost its forest	UMD Hansen GFC; EU JRC TMF	30 m	annual	1990 / 2001 → 2025	$\approx 30 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	A
burned500	date each pixel burned	NASA MODIS MCD64A1	500 m	monthly	2000-11 →	$\approx 10 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	A
cat_gfm	list of flood and surface-water maps	Copernicus GFM; NASA OPERA DSWx	20–30 m	per pass	2015 / 2016 →	$< 1 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	A
biomass100	yearly above-ground biomass	ESA CCI Biomass; GEDI L4B	100 m (GEDI 1 km)	annual	2005–2012, 2015–2024	$\approx 0.4 \text{ TB}$	<i>inherited from 1.0.tf</i> ; public	A (2020 and GEDI L4B) / B
landcover10	yearly land-cover and crop-type maps	Esri/IO, ESA, MapBiomass, USDA, JRC	10–30 m	annual	1985 / 2017 →	$\approx 0.2 \text{ TB}^\dagger$	<i>inherited from 1.0.tf</i> ; public	C
static_fine	fixed maps: elevation and depth, land cover, distance to coast	GEBCO, ESA WorldCover, GSHHG	$15''$ ($\approx 460 \text{ m}$); 0.05°	fixed	—	$\approx 8 \text{ GB}$	<i>inherited from 1.0.tf</i> ; public	A

Track and phase of an inherited store are 1.0.tf’s. The ten image catalogues of 1.0.tf (**cat_landsat**, **cat_hls**, **cat_s2**, **cat_s1**, **cat_nisar**, **cat_palsar**, **cat_olci**, **cat_viirs**, **cat_ecostress**, **cat_submetre**), the tile tokens and **oc300_coast** are **not** inherited: the four AlphaEarth stores replace them. Inherited sizes and probes are 1.0.tf’s (**Probe**: a one-month test fetch, E-082); † estimate; B = bytes.

How it is stored. The registry **tensors/family09_tf/family09tf.json** carries an **inherits** block naming family 1.0.tf’s registry; each inherited store keeps its 1.0.tf path and checksums (the four private ones stay in the private repository), and only the four AlphaEarth groups live under **family09_tf**. The pooled stores (**aef1k**, **aef100r**) average the unit-length 10 m vectors in each cell, renormalise, and keep the average’s length as a 65th *coherence* channel (1 = the cell is one thing, near 0 = many), all in one byte per number, as compressed tiles; **aef_dots** keeps raw vectors as point rows; **aef10** points at Google’s files. A year- Y vector is filed in the first five-day bin after 31 December Y , so a prediction inside year Y never sees that year’s summary.

The ladder

- A** ≈ 0.2 TB: `aef1k`, `aef_dots`, the `aef10` reference and the registry with its `inherits` block; built in one pass next to the mirror, one lane per year, \$50–150. Inherited stores appear as 1.0.tf publishes them (its phases: A ≈ 0.7 TB, B 2.3–2.7 TB, C ≈ 3 TB).
- B** `aef100r` over three deforestation frontiers (Amazon, Congo, insular Southeast Asia), ≈ 0.35 TB.
- C** `aef100r` over the ± 100 km coastal band (≈ 5.7 TB), once E-078c's coastal cells exist to compare against.

Glossary

Every term, acronym, producer, instrument and code name used in this summary and in the full design note, in alphabetical order.

aef10 — Family 0.9.tf registry group pointing at Google’s original 10 m files, read in place.

aef100r — Family 0.9.tf store of the embedding averaged over ≈ 100 m cells in chosen regions.

aef1k — Family 0.9.tf store of the embedding averaged over each ≈ 1 km cell, with a coherence channel.

AEF band names A00–A63 — The names of the 64 embedding values in Earth Engine.

aef_custom — A possible private store of sub-annual AlphaEarth embeddings, only if Google’s academic programme grants a sample.

aef_dots — Family 0.9.tf store of 16 unaveraged 10 m embedding vectors per 0.25° land cell per year, at the same positions every year.

aef_index.parquet — Google’s 70 MB table of every embedding tile.

alerts — Family 1.0.tf target store of dated forest-disturbance alerts (GLAD, RADD, OPERA) as point rows.

ALOS, PALSAR, PALSAR-2 — Japan’s Advanced Land Observing Satellites and their L-band radars.

AlphaEarth Foundations (AEF) — Google DeepMind’s model that turns sequences of satellite images into one 64-number description of each 10 m of land for each year; its published output is the *Satellite Embedding*.

ALTICAP — A coastal altimetry product that reprocesses Jason-3 to within a few hundred metres of the shore.

alticap — Family 1.0.tf store of ALTICAP coastal sea level along Jason-3 tracks, 2016–2021.

altimeter, along-track — A radar satellite that measures sea-surface height straight below it; along-track data are its measurements along that ground line.

AmeriFlux, ICOS, TERN, ChinaFLUX, FLUXNET — The flux-tower networks of the Americas, Europe, Australia and China, and FLUXNET, their global collection (the FLUXNET Shuttle release).

anchor — The place and time the model is asked to predict; everything it reads is placed relative to it.

annual target — A yearly map the model is asked to predict (a loss year, a biomass map); it is filed at the end of its year so it is never read as an input from inside that year.

attribution, yes, no (licence.redistribution) — Registry values saying whether a licence lets the data be republished freely, with credit, or not at all.

AVISO, ODATIS — French services that distribute satellite altimetry and ocean data.

AWS, AWS Open Data — Amazon Web Services, and its programme that hosts public datasets at no cost to readers.

bin (pentad) — The five-day time step every family uses, counted from 1 January 1982; a bin before 1982 has a negative number.

BIOMASS (the satellite), P band — ESA’s radar satellite, launched 29 April 2025; its 70 cm wavelength reaches through a canopy to the trunks, so it measures woody biomass directly. **cat_biomass_esa** catalogues its products.

biomass100 — Family 1.0.tf target store of yearly above-ground biomass maps at 100 m.

box (rented, verified) — A machine hired by the hour from a marketplace; it holds no credentials, and a verified one runs on a host the marketplace has vetted.

B/row (bytes per row) — Stored size of one row of a point store; a row costs $27 + 2$ bytes per channel ($31 + 2$ per channel with the 64-bit time).

burned500 — Family 1.0.tf target store of the date each 500 m pixel burned.

canopy30 — Family 1.0.tf target store: GLAD’s 2020 forest-height map at 30 m, the canopy read-out’s annual target.

canopy_ref — Family 1.0.tf catalogue of the finer canopy-height maps, ETH Zurich’s 10 m and Meta/WRI’s 1 m, listed with their download links.

Caravan — An open (CC BY 4.0) collection of catchment discharge and weather data, including a GRDC subset.

cat_biomass_esa — Family 1.0.tf catalogue of the released products of ESA’s BIOMASS radar satellite.

cat_ecostress — Catalogue of ECOSTRESS 70 m thermal scenes.

cat_gfm — Target catalogue of flood and surface-water maps (Copernicus GFM, OPERA DSWx); a read-out, not imagery.

cat_hls — Catalogue of NASA’s harmonised Landsat and Sentinel-2 scenes.

cat_landsat — Catalogue of every Landsat scene since 1982, all sensors.

cat_nisar — Catalogue of NISAR L-band radar scenes.

cat_olci — Catalogue of Sentinel-3 OLCI 300 m ocean-colour scenes over the coastal band.

cat_palsar — Catalogue of JAXA’s PALSAR yearly radar mosaics and wide-swath scenes (exception E5).

cat_s1 — Catalogue of Sentinel-1 radar scenes.

cat_s2 — Catalogue of Sentinel-2 optical scenes.

cat_submetre — Catalogue of sub-metre and metre-scale aerial and basemap imagery where it is openly published (exception E6).

cat_viirs — Catalogue of VIIRS 375 m imager scenes.

C band, L band — Radar wavelengths of about 5.5 cm and 24 cm; L band sees deeper into vegetation.

CC0, CC BY, CC BY-NC — Creative Commons licences: public domain; free with attribution; free with attribution for non-commercial use only.

C (channels) — The number of values stored per row or per pixel.

checksum (sha256) — A 64-character fingerprint of a file; if one byte changes, the fingerprint changes.

chfrank/earth-tensors, chfrank/earth-tensors-private — The project’s public and private Hugging Face dataset repositories.

CHIRPS — The Climate Hazards Center InfraRed Precipitation with Station data rainfall record (from 1981).

chirps05 — Family 1.0.tf store of CHIRPS five-day rainfall at 0.05° (exception E4), with its **frame_table** of frame dates.

CMEMS, CV, UZ — Three GESLA contributor codes whose tide-gauge data are for research use only.

codec (local codec) — A small learned encoder that turns an image tile into a few compact vectors (tokens) the model can read.

COG — Cloud-optimised GeoTIFF: an image file laid out so any part of it can be read by byte range over HTTP.

coherence — In **aef1k**, the length of the average of a cell’s unit vectors: 1 if its pixels agree, near 0 if they differ.

companding (int8) — Storing a value in one signed byte on a square-root scale, which keeps more precision near zero; AlphaEarth’s files use it.

cone — The set of past measurements the model reads around the anchor: wider for older times, like a cone in space and time.

Copernicus, Copernicus Marine — The EU’s Earth-observation programme and its marine service (free account required).

Copernicus DEM, GLO-30 — The EU’s global 30 m elevation model.

Custom Satellite Embeddings — Google Maps Platform’s sub-annual embeddings, in private preview with no public licence.

DAAC (LP, ORNL, PO, OB) — NASA’s discipline data centres: LP DAAC for land, ORNL DAAC for ecology, PO.DAAC for oceans, OB.DAAC for ocean biology.

DAC, GDAC — Data assembly centre, and the global one that gathers a network’s data (Argo, gliders, OceanSITES).

DeepMind — Google’s AI research laboratory, co-producer of AlphaEarth.

distribution — Registry field naming a store’s track, **public** or **private**.

done.json — A marker a build writes last, so an unfinished build can never look finished.

DSWx-HLS, DSWx-S1 — OPERA’s 30 m surface-water extent maps from HLS optical and from Sentinel-1 radar.

E-078, E-078c — The design study that gave family 10 one schema for every granularity, and its first experiment training a local image codec on about 300 coastal cells.

E-082 — The build plan for families 1.gf, 1.0.tf and 0.9.tf; its first wave wrote and probed the phase-A point-store adapters.

E1–E6 (exceptions) — Sources admitted although they break the 10 km / 5-day rule: E1 scatterometer wind at 12.5 km, E2 IMERG rain at 11 km, E3 GOSAT at 10.5 km, E4 CHIRPS rain, E5 PALSAR yearly mosaics, E6 sub-metre photos.

Earth Engine — Google’s hosted geospatial analysis platform (account required).

ecad — Private family 1.0.tf store of ECA&D European station series missing from GHCN-Daily.

ECA&D; TX, TN, TG, RR, SS, SD, HU, PP — The European Climate Assessment & Dataset of station series, and its element codes: maximum, minimum and mean temperature, rain, sunshine, snow depth, humidity and pressure.

ECMWF — The European Centre for Medium-Range Weather Forecasts, producer of ERA5.

embedding — A learned list of numbers summarising what was observed at a place, arranged so that similar places get similar numbers.

ERA5, ERA5-Land — ECMWF’s global reanalysis and its land-surface companion.

ESA — The European Space Agency.

ESA CCI — ESA’s Climate Change Initiative, which builds long consistent climate records from satellites.

Esri, Impact Observatory — The companies that produce a yearly 10 m global land-cover map.

falsifier — The test, fixed before building, that can show a store is not worth keeping.

family — A numbered set of datasets built to one storage design and read by the programme’s forecasting model.

family 0.9.tf — Family 1.0.tf with its raw-imagery catalogue replaced by Google’s AlphaEarth embedding, everything else inherited.

family 10, 10.1, 10.2 — The granularity-aware dataset set in which every measurement keeps its own place, time and footprint; 10.1 built four point stores (drifters, tropical moorings, ship CO₂, satellite sea level) and 10.2 inherited them and added fishing effort.

family10.json, family1gf.json, family1tf.json, family09tf.json — The registry files of families 10, 1.gf, 1.0.tf and 0.9.tf.

family 1.0.tf — The proposed land-and-coast family at the same scale, built from raw observations and a catalogue of raw imagery (“tf”: terrestrial, fine).

family 1.gf — The proposed global family of ocean and atmosphere measurements finer than 10 km and 5 days (“gf”: global, fine).

family 7, 7.1, 7.2 — The published global input tensor: every 0.25° cell from pole to pole in five-day steps, 1982–2024; 7.1 added ocean colour and 7.2 is the current version.

fire — Family 1.0.tf store of FIRMS satellite fire detections.

FIRMS; FRP — NASA’s Fire Information for Resource Management System, and fire radiative power, the energy a fire gives off.

flux — Family 1.0.tf store of half-hourly flux-tower measurements.

footprint; $\log_2 f_p$, $\log_2 f_t$ ($\log2_fp$, $\log2_dt$, fp) — The area and time span a value averages over, stored as base-2 logarithms relative to 27.83 km and 5 days; -4 is the label for a point measurement with no area.

frame; frames_per_bin, frame_seconds, F — One time slice of a gridded field; a five-day bin holds F of them (5 for daily, 40 for three-hourly), each lasting **frame_seconds**.

GBIF — The Global Biodiversity Information Facility, an open database of species occurrences.

GCS ($gs://$) — Google Cloud Storage, Google’s object store; **$gs://$** addresses a file in it.

GDAL, VRT — GDAL is the standard open library for geospatial rasters; a VRT is its small text file that describes how to read another raster.

gbif — Family 1.0.tf store of dated, located species-occurrence records from GBIF’s monthly snapshot.

GBIF — The Global Biodiversity Information Facility, the pooled open archive of species-occurrence records — and one of AlphaEarth’s own training targets.

GEBCO — The General Bathymetric Chart of the Oceans, a global elevation and sea-floor-depth grid.

gedi — Family 1.0.tf store of GEDI laser footprints: canopy height profile, cover and biomass.

GEDI; L2A, L2B, L4A, L4B — The Global Ecosystem Dynamics Investigation, a laser on the Space Station measuring canopy height; its products for height profiles, cover, footprint biomass and gridded biomass.

GESLA, GESLA-4 — The Global Extreme Sea Level Analysis compilation of hourly tide-gauge records, version 4.

GFM — Copernicus’ Global Flood Monitoring service, flood extent from every Sentinel-1 pass.

ghcnd — Family 1.0.tf store of GHCN-Daily station records.

GHCN-Daily, GHcNh, ISD — NOAA’s Global Historical Climatology Network of daily and hourly station records; GHcNh replaces the frozen Integrated Surface Database.

ghcnh — Family 1.0.tf store of GHCN-hourly station reports.

GLAD, GLAD-L, GLAD-S2 — The University of Maryland’s forest-loss alerts from Landsat and from Sentinel-2.

gliders — Family 1.0.tf store of glider profiles at 16 depths.

GRACE — Satellites that measure changes in Earth’s gravity, and so in stored water and ice.

GRDC — The Global Runoff Data Centre’s archive of river discharge from more than 10,000 gauges.

GSHHG — The Global Self-consistent Hierarchical High-resolution Geography shoreline set (public domain).

Hansen GFC — The University of Maryland’s Global Forest Change map of tree cover in 2000 and the year of loss since.

hfr — Family 1.0.tf store of hourly surface currents from shore-based radar.

HFR, HFRNet — High-frequency radar that measures surface currents from the shore, and the US network of such radars (with a European node).

HTTP, HTTPS, HTTP range — The web’s transfer protocol (HTTPS is the encrypted form); a range request asks for one byte span of a file.

Hub (Hugging Face) — Hugging Face’s dataset hosting, where the project publishes its stores; it limits files per repository and per folder.

ICESat-2 — NASA’s polar laser-altimetry satellite (2018 on).

icesat2 — Family 1.0.tf store of ICESat-2 terrain, canopy and sea-ice heights per 100 m segment.

IGRA — The Integrated Global Radiosonde Archive of weather-balloon soundings.

igra — Family 1.0.tf store of IGRA radiosonde soundings at 16 standard levels.

IMS — NOAA’s Interactive Multisensor Snow and Ice Mapping System, a daily analyst-drawn chart.

ims1k — Family 1.0.tf store of the daily 1 km IMS snow and ice chart.

inherits — Registry block naming another family’s registry and the groups taken from it unchanged, each with its original path and checksums.

IOOS — The US Integrated Ocean Observing System, which runs the national glider data centre.

ISMN — The International Soil Moisture Network of in-ground sensors.

ismn — Private family 1.0.tf store of ISMN soil moisture and soil temperature.

ISRO — The Indian Space Research Organisation, NASA’s partner on NISAR.

Jason-3 — A radar altimeter satellite (2016 on) measuring sea level.

JAXA — Japan’s space agency.

JRC — The European Commission’s Joint Research Centre.

JRC TMF — The JRC’s Tropical Moist Forest map of yearly forest change since 1990.

KB, MB, GB, TB, PB — Kilo-, mega-, giga-, tera- and petabytes: each a thousand times the last.

KNMI — The Royal Netherlands Meteorological Institute, which runs ECA&D.

lai500 — Family 1.0.tf store of eight-day leaf area and absorbed light at 500 m from MODIS and VIIRS (exception E7).

landcover10 — Family 1.0.tf target store of yearly land-cover and crop-type maps.

Landsat 4, 5, 7, 8, 9; TM, ETM+, OLI, TIRS — The US optical Earth-observation satellites since 1982, and their cameras: Thematic Mapper, Enhanced Thematic Mapper Plus, Operational Land Imager and Thermal Infrared Sensor; Collection 2 is the current reprocessing.

lat, lon, values, qc, platform, time_s — Arrays of a point store: position, the measured values, a quality code, a hashed instrument identity, and the time in seconds since 1982.

layout, prefix, path, cadence, footprint, tier — Registry fields per group: storage layout, folder, the group’s own location, time step, footprint and tier.

LiDAR — Laser ranging: a pulse’s return time gives the height of the ground and of each layer of a canopy.

leaf-area index (LAI), FPAR — The area of leaf above a square metre of ground, and the fraction of usable sunlight the canopy absorbs.

lossyear — Family 1.0.tf target store of the year each 30 m pixel lost its forest.

1st05 — Family 1.0.tf store of daily 0.05° land-surface temperature, day and night.

Major-TOM — An open Earth-observation dataset project whose Hugging Face sample of the AlphaEarth embedding is used as a test fixture.

MapBiomass — Brazil’s annual 30 m land-cover mapping project.

MOD09CMG, MOD10C1, MOD11C1, MYD11C1, MCD64A1 — MODIS product names: daily 0.05° reflectance, snow cover and land-surface temperature (MOD for Terra, MYD for Aqua), and monthly burned area (MCD, both).

MODIS; Terra, Aqua — The Moderate Resolution Imaging Spectroradiometer on NASA’s Terra (1999) and Aqua (2002) satellites, both near the end of their lives.

NaN — “Not a number”, the stored value for “not measured”.

NASA — The US space agency.

NCEI, OCADS — NOAA’s National Centers for Environmental Information, the US national environmental archive, and its ocean-carbon archive.

NDBC — NOAA’s National Data Buoy Center.

ndbc — Family 1.0.tf store of NDBC buoy and coastal-station weather and waves.

NISAR; GCOV — The NASA-ISRO L-band radar satellite (launched 2025), and its geocoded covariance product.

NLCD — The US National Land Cover Database.

NOAA, NOAA STAR — The US National Oceanic and Atmospheric Administration, and its satellite research centre.

nodata — The marker value a file uses for “no measurement here”.

n_R — The number of readings a token carries; $n_R = 0$ says “nothing was measured here”.

NSIDC — The US National Snow and Ice Data Center.

NWIS — The USGS National Water Information System, the US river-gauge archive.

oc300_coast — A possible phase-C gridded store of 300 m OLCI ocean colour over the coastal band.

OceanGliders — The international glider programme and its data centre.

OPERA — NASA’s Observational Products for End-Users from Remote Sensing Analysis: disturbance and surface-water maps from HLS and Sentinel-1.

orthophoto — An aerial photograph corrected so that it lines up with a map.

phase A, B, C — The three build waves of each family: A cheap and first, B the terabyte stores, C decisions taken after A and B have verdicts.

pheno500 — Family 1.0.tf target store of the yearly leaf calendar at 500 m: green-up, peak, senescence and dormancy dates.

phenology — The calendar of a plant’s year — when it leafs out, peaks, dies back and goes dormant.

probe — A one-month (or one-zone) test run of a store’s real fetch code, measuring rows, bytes and speed before any full build.

public track, private track — Where a store lives: the public Hugging Face repository that anyone can read, or a private one used only by this project because the source’s licence forbids republishing.

RADD — Radar for Detecting Deforestation, Wageningen’s Sentinel-1 forest-loss alerts.

radiosonde — An instrument package on a weather balloon that measures temperature, humidity and wind as it rises.

read-out — A quantity the model is asked to predict or report, such as the Niño-3.4 index or next year’s forest loss.

reanalysis — A weather or ocean model run over past decades and continually corrected towards observations, giving complete fields that are partly model.

ref105 — Family 1.0.tf store of daily 0.05° land-surface reflectance.

registry — The JSON file listing every store of a family with its tier, location, channels, checksums and licence; the one file a reader opens first.

river — Family 1.0.tf store of public daily river discharge (USGS and Caravan).

river_grdc — Private family 1.0.tf store of GRDC daily river discharge.

S3 (s3://) — Amazon’s object-storage service and its address form; many archives serve files through it.

SANHO — The South African Navy Hydrographic Office.

schema 2, schema 3 — The column layout of a point store: schema 3 widens the time column to 64 bits so records before 1914 fit.

seed — One independent training run; three seeds means the same test run three times with different random starts.

Sentinel-1, -2, -3, -5P, -6A (S1A, S1C, S1D, S2A, S2B, S2C) — The EU Copernicus satellites: C-band radar, 10 m optical, ocean colour and temperature, trace gases, and sea-level altimetry; the letters are individual satellites of a series.

shard, tile; shard_index.npy, tile_grid.json, n_tiles — In the G-fine layout, one compressed file per group and bin, cut into 256×256 -pixel tiles; the index gives each tile's byte position and the grid file each tile's corners.

siblings — A list in family 10.2's registry naming the registries of families 1.gf, 1.0.tf and 0.9.tf.

slatrack — Family 10.1's store of satellite sea level along the ground track, all missions (67 GB).

snow05 — Family 1.0.tf store of daily 0.05° snow cover.

Source Cooperative (tge-labs/aef) — A public data host where the Taylor Geospatial Engine keeps a complete mirror of the AlphaEarth embedding.

spot instance — A discounted cloud machine that the provider may reclaim at short notice.

sif — Family 1.0.tf store of solar-induced chlorophyll fluorescence soundings from TROPOMI, OCO-2 and OCO-3.

SIF (solar-induced chlorophyll fluorescence) — The faint red light a leaf re-emits while photosynthesising — the closest thing to a direct measurement of photosynthesis from orbit.

SST — Sea-surface temperature.

static_fine — Family 1.0.tf store of fixed fields: elevation and bathymetry, land-cover fractions, distance to coast.

store — One dataset of a family, published as a small set of arrays plus a description file.

support period, valid period — The dates of the images an AlphaEarth vector was computed from, and the period it describes (one calendar year in the published product).

tide — Family 1.0.tf store of public GESLA tide-gauge sea level.

tide_private — Private sibling of **tide** holding the gauges whose terms forbid republishing.

tier G-fine (sharded) — The layout family 1.gf adds for grids finer than 0.25° : each five-day bin is one compressed file cut into tiles, so a read fetches only the tiles it needs.

tier G, P, T — Family 10's three storage forms: G, gridded fields read one time step at a time; P, points, profiles and tracks read by nearest-neighbour search; T, image tiles held as a catalogue plus codec tokens.

tile tokens — The codec's vectors for each image tile, a later store sized by E-078c.

token — The unit the model reads: a value, its offset in space and time from the anchor, and its footprint.

TROPOMI — The trace-gas imager on Sentinel-5P, measuring NO_2 , CO and CH_4 (nitrogen dioxide, carbon monoxide, methane).

TU Wien — The Technical University of Vienna, which hosts ISMN.

UCSB CHC — The Climate Hazards Center at the University of California, Santa Barbara, producer of CHIRPS.

UHSLC — The University of Hawaii Sea Level Center.

UK, US, EU — United Kingdom, United States, European Union.

UMD — The University of Maryland, producer of GLAD and the Hansen forest-change map.

unit length, S^{63} — AlphaEarth's vectors all have length one, so they lie on a sphere in 64 dimensions and their dot product measures similarity.

URL — A web address.

USDA — The US Department of Agriculture.

USGS — The US Geological Survey.

UTC — Coordinated Universal Time, the time zone of weather observations.

UTM — Universal Transverse Mercator: sixty map zones that each keep a flat metre grid; AlphaEarth's tiles use them.

VIIRS; S-NPP, NOAA-20, NOAA-21 — The Visible Infrared Imaging Radiometer Suite, MODIS's successor, on the Suomi-NPP (ending 2026-11), NOAA-20 and NOAA-21 satellites; I-band is its 375 m channel set.

WorldCover — ESA's 10 m global land-cover map (2020 and 2021).

WUR — Wageningen University & Research, producer of RADD.